ScholarGraph • AI Literature Synthesis & Paper Recommendations

Dual-Signal AI Retrieval (SciBERT Semantics + GraphSAGE Topology) • Google Gemini LLM Synthesis

Active Research Anchor (Seed Paper)

Title: How Powerful are Graph Neural Networks?

Year: 2018 • **Citations:** 10,562 • **Paper ID:** 62ed9bf1d83c8db1...

Abstract: *Graph Neural Networks (GNNs) are an effective framework for representation learning of graphs. GNNs follow a neighborhood aggregation scheme, where the representation vector of a node is computed by recursively aggregating and transforming representation vectors of its neighboring nodes. Many GNN variants have been proposed and have achieved state-of-the-art results on both node and graph classification tasks. However, despite GNNs revolutionizing graph representation learning, there is limited understanding of their representational properties and limitations. Here, we present a theoretical framework for analyzing the expressive power of GNNs to capture different graph structures. Our results characterize the discriminative power of popular GNN variants, such as Graph Convolutional Networks and GraphSAGE, and show that they cannot learn to distinguish certain simple graph structures. We then develop a simple architecture that is provably the most expressive among the class of GNNs and is as powerful as the Weisfeiler-Lehman graph isomorphism test. We empirically validate our theoretical findings on a number of graph classification benchmarks, and demonstrate that our model achieves state-of-the-art performance.*

AI Research Advisor Synthesis (Google Gemini 3.5 Flash-Lite)

Google Gemini Scholarly Synthesis & Reading Pathway:

These foundational papers collectively establish the scaling, inductive generalization, and hierarchical structuring mechanisms necessary to evaluate the expressive limits and representation power of Graph Neural Networks (GNNs). You should read **'Inductive Representation Learning on Large Graphs' (GraphSAGE, 2017)** first because its neighborhood-sampling framework provides the core algorithmic baseline for understanding how GNNs aggregate localized topological information and generalize to unseen nodes.

LangGraph Agentic Execution Pipeline (4-Stage State Machine)

Pipeline Stage	Agent Node	Action & Outcome	Duration
1. Intent Analysis Agent	IntentAnalyzerNode	Classified search intent as 'seed_exploration'	**0.3 ms**
2. Semantic Retrieval Agent	SemanticRetrieverNode	SciBERT Deep Vector Search	**21.78 ms**
3. GraphSAGE Fusion Agent	GraphSAGEFusionNode	Topological Link Affinity & Dual Signal Fusion	**1.07 ms**
4. Explainability & Synthesis Agent	ExplainabilityAgentNode (Google Gemini: gemini-3.5-flash-lite)	Generated Gemini LLM Literature Synthesis & Rationales	**1226.27 ms**

Top Recommended Papers (Ranked by SciBERT + GraphSAGE Fusion, $\alpha = 0.60$)

#1 Hierarchical Graph Representation Learning with Differentiable Pooling

100% MATCH

Year: 2018 • **Citations:** 2,612 • **Semantics:** 95% • **Citation Graph:** 99% • *Dual-signal convergence: 95.0% conceptual similarity and 98.8% citation neighborhood proximity.*

Abstract: Recently, graph neural networks (GNNs) have revolutionized the field of graph representation learning through effectively learned node embeddings, and achieved state-of-the-art results in tasks such as node classification and link prediction. However, current GNN methods are inherently flat and do not learn hierarchical representations of graphs---a limitation that is especially problematic for the task of graph classification, where the goal is to predict the label associated with an entire graph. Here we propose DiffPool, a differentiable graph pooling module that can generate hierarchical representations of graphs and can be combined with various graph neural network architectures in an end-to-end fashion. DiffPool learns a differentiable soft cluster assignment for nodes at each layer of a deep GNN, mapping nodes to a set of clusters, which then form the coarsened input for the next GNN layer. Our experimental results show that combining existing GNN methods with DiffPool yields an average improvement of 5-10% accuracy on graph classification benchmarks, compared to all existing pooling approaches, achieving a new state-of-the-art on four out of five benchmark datasets.

#2 Inductive Representation Learning on Large Graphs

99% MATCH

Year: 2017 • **Citations:** 21,376 • **Semantics:** 94% • **Citation Graph:** 97% • *Dual-signal convergence: 94.3% conceptual similarity and 97.3% citation neighborhood proximity.*

Abstract: Low-dimensional embeddings of nodes in large graphs have proved extremely useful in a variety of prediction tasks, from content recommendation to identifying protein functions. However, most existing approaches require that all nodes in the graph are present during training of the embeddings; these previous approaches are inherently transductive and do not naturally generalize to unseen nodes. Here we present GraphSAGE, a general, inductive framework that leverages node feature information (e.g., text attributes) to efficiently generate node embeddings for previously unseen data. Instead of training individual embeddings for each node, we learn a function that generates embeddings by sampling and aggregating features from a node's local neighborhood. Our algorithm outperforms strong baselines on three inductive node-classification benchmarks: we classify the category of unseen nodes in evolving information graphs based on citation and Reddit post data, and we show that our algorithm generalizes to completely unseen graphs using a multi-graph dataset of protein-protein interactions.

#3 Graph Convolutional Neural Networks for Web-Scale Recommender Systems

99% MATCH

Year: 2018 • **Citations:** 4,315 • **Semantics:** 95% • **Citation Graph:** 98% • *Dual-signal convergence: 94.9% conceptual similarity and 97.6% citation neighborhood proximity.*

Abstract: Recent advancements in deep neural networks for graph-structured data have led to state-of-the-art performance on recommender system benchmarks. However, making these methods practical and scalable to web-scale recommendation tasks with billions of items and hundreds of millions of users remains an unsolved challenge. Here we describe a large-scale deep recommendation engine that we developed and deployed at Pinterest. We develop a data-efficient Graph Convolutional Network (GCN) algorithm, which combines efficient random walks and graph convolutions to generate embeddings of nodes (i.e., items) that incorporate both graph structure as well as node feature information. Compared to prior GCN approaches, we develop a novel method based on highly efficient random walks to structure the convolutions and design a novel training strategy that relies on harder-and-harder training examples to improve robustness and convergence of the model. We also develop an efficient MapReduce model inference algorithm to generate embeddings using a trained model. Overall, we can train on and embed graphs that are four orders of magnitude larger than typical GCN implementations. We show how GCN embeddings can be used to make high-quality recommendations in various settings at Pinterest, which has a massive underlying graph with 3 billion nodes representing pins and boards, and 17 billion edges. According to offline metrics, user studies, as well as A/B tests, our approach generates higher-quality recommendations than comparable deep learning based systems. To our knowledge, this is by far the largest application of deep graph embeddings to date and paves the way for a new generation of web-scale recommender systems based on graph convolutional architectures.

#4 Geom-GCN: Geometric Graph Convolutional Networks

99% MATCH

Year: 2020 • **Citations:** 1,568 • **Semantics:** 95% • **Citation Graph:** 97% • *Dual-signal convergence: 95.2% conceptual similarity and 97.4% citation neighborhood proximity.*

Abstract: Message-passing neural networks (MPNNs) have been successfully applied in a wide variety of applications in the real world. However, two fundamental weaknesses of MPNNs' aggregators limit their ability to represent graph-structured data: losing the structural information of nodes in neighborhoods and lacking the ability to capture long-range dependencies in disassortative graphs. Few studies have noticed the weaknesses from different perspectives. From the observations on classical neural network and network geometry, we propose a novel geometric aggregation scheme for graph neural networks to overcome the two weaknesses. The behind basic idea is the aggregation on a graph can benefit from a continuous space underlying the graph. The proposed aggregation scheme is permutation-invariant and consists of three modules, node embedding, structural neighborhood, and bi-level aggregation. We also present an implementation of the scheme in graph convolutional networks, termed Geom-GCN, to perform transductive learning on graphs. Experimental results show the proposed Geom-GCN achieved state-of-the-art performance on a wide range of open datasets of graphs.

#5 Representation Learning on Graphs with Jumping Knowledge Networks

99% MATCH

Year: 2018 • **Citations:** 2,473 • **Semantics:** 94% • **Citation Graph:** 98% • *Dual-signal convergence: 94.5% conceptual similarity and 97.6% citation neighborhood proximity.*

Abstract: Recent deep learning approaches for representation learning on graphs follow a neighborhood aggregation procedure. We analyze some important properties of these models, and propose a strategy to overcome those. In particular, the range of "neighboring" nodes that a node's representation draws from strongly depends on the graph structure, analogous to the spread of a random walk. To adapt to local neighborhood properties and tasks, we explore an architecture -- jumping knowledge (JK) networks -- that flexibly leverages, for each node, different neighborhood ranges to enable better structure-aware representation. In a number of experiments on social, bioinformatics and citation networks, we demonstrate that our model achieves state-of-the-art performance. Furthermore, combining the JK framework with models like Graph Convolutional Networks, GraphSAGE and Graph Attention Networks consistently improves those models' performance.

#6 node2vec: Scalable Feature Learning for Networks

99% MATCH

Year: 2016 • **Citations:** 12,642 • **Semantics:** 95% • **Citation Graph:** 95% • *Dual-signal convergence: 95.3% conceptual similarity and 94.7% citation neighborhood proximity.*

Abstract: Prediction tasks over nodes and edges in networks require careful effort in engineering features used by learning algorithms. Recent research in the broader field of representation learning has led to significant progress in automating prediction by learning the features themselves. However, present feature learning approaches are not expressive enough to capture the diversity of connectivity patterns observed in networks. Here we propose node2vec, an algorithmic framework for learning continuous feature representations for nodes in networks. In node2vec, we learn a mapping of nodes to a low-dimensional space of features that maximizes the likelihood of preserving network neighborhoods of nodes. We define a flexible notion of a node's network neighborhood and design a biased random walk procedure, which efficiently explores diverse neighborhoods. Our algorithm generalizes prior work which is based on rigid notions of network neighborhoods, and we argue that the added flexibility in exploring neighborhoods is the key to learning richer representations. We demonstrate the efficacy of node2vec over existing state-of-the-art techniques on multi-label classification and link prediction in several real-world networks from diverse domains. Taken together, our work represents a new way for efficiently learning state-of-the-art task-independent representations in complex networks.

#7 DeepFool: A Simple and Accurate Method to Fool Deep Neural Networks

99% MATCH

Year: 2015 • **Citations:** 5,550 • **Semantics:** 94% • **Citation Graph:** 97% • *Dual-signal convergence: 94.3% conceptual similarity and 97.2% citation neighborhood proximity.*

Abstract: State-of-the-art deep neural networks have achieved impressive results on many image classification tasks. However, these same architectures have been shown to be unstable to small, well sought, perturbations of the images. Despite the importance of this phenomenon, no effective methods have been proposed to accurately compute the robustness of state-of-the-art deep classifiers to such perturbations on large-scale datasets. In this paper, we fill this gap and propose the DeepFool algorithm to efficiently compute perturbations that fool deep networks, and thus reliably quantify the robustness of these classifiers. Extensive experimental results show that our approach outperforms recent methods in the task of computing adversarial perturbations and making classifiers more robust.

#8 Network Dissection: Quantifying Interpretability of Deep Visual Representations

99% MATCH

Year: 2017 • **Citations:** 1,847 • **Semantics:** 94% • **Citation Graph:** 98% • *Dual-signal convergence: 94.4% conceptual similarity and 97.7% citation neighborhood proximity.*

Abstract: We propose a general framework called Network Dissection for quantifying the interpretability of latent representations of CNNs by evaluating the alignment between individual hidden units and a set of semantic concepts. Given any CNN model, the proposed method draws on a data set of concepts to score the semantics of hidden units at each intermediate convolutional layer. The units with semantics are labeled across a broad range of visual concepts including objects, parts, scenes, textures, materials, and colors. We use the proposed method to test the hypothesis that interpretability is an axis-independent property of the representation space, then we apply the method to compare the latent representations of various networks when trained to solve different classification problems. We further analyze the effect of training iterations, compare networks trained with different initializations, and measure the effect of dropout and batch normalization on the interpretability of deep visual representations. We demonstrate that the proposed method can shed light on characteristics of CNN models and training methods that go beyond measurements of their discriminative power.

#9 Composition-based Multi-Relational Graph Convolutional Networks

99% MATCH

Year: 2019 • **Citations:** 1,209 • **Semantics:** 95% • **Citation Graph:** 96% • *Dual-signal convergence: 95.4% conceptual similarity and 96.4% citation neighborhood proximity.*

Abstract: Graph Convolutional Networks (GCNs) have recently been shown to be quite successful in modeling graph-structured data. However, the primary focus has been on handling simple undirected graphs. Multi-relational graphs are a more general and prevalent form of graphs where each edge has a label and direction associated with it. Most of the existing approaches to handle such graphs suffer from over-parameterization and are restricted to learning representations of nodes only. In this paper, we propose CompGCN, a novel Graph Convolutional framework which jointly embeds both nodes and relations in a relational graph. CompGCN leverages a variety of entity-relation composition operations from Knowledge Graph Embedding techniques and scales with the number of relations. It also generalizes several of the existing multi-relational GCN methods. We evaluate our proposed method on multiple tasks such as node classification, link prediction, and graph classification, and achieve demonstrably superior results. We make the source code of CompGCN available to foster reproducible research.

#10 Deep Gaussian Embedding of Graphs: Unsupervised Inductive Learning via Ranking

98% MATCH

Year: 2017 • **Citations:** 799 • **Semantics:** 94% • **Citation Graph:** 98% • *Dual-signal convergence: 94.3% conceptual similarity and 98.1% citation neighborhood proximity.*

Abstract: Methods that learn representations of nodes in a graph play a critical role in network analysis since they enable many downstream learning tasks. We propose Graph2Gauss - an approach that can efficiently learn versatile node embeddings on large scale (attributed) graphs that show strong performance on tasks such as link prediction and node classification. Unlike most approaches that represent nodes as point vectors in a low-dimensional continuous space, we embed each node as a Gaussian distribution, allowing us to capture uncertainty about the representation. Furthermore, we propose an unsupervised method that handles inductive learning scenarios and is applicable to different types of graphs: plain/attributed, directed/undirected. By leveraging both the network structure and the associated node attributes, we are able to generalize to unseen nodes without additional training. To learn the embeddings we adopt a personalized ranking formulation w.r.t. the node distances that exploits the natural ordering of the nodes imposed by the network structure. Experiments on real world networks demonstrate the high performance of our approach, outperforming state-of-the-art network embedding methods on several different tasks. Additionally, we demonstrate the benefits of modeling uncertainty - by analyzing it we can estimate neighborhood diversity and detect the intrinsic latent dimensionality of a graph.